

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	Shaik Mohammad Sohan	Reg. No.:	192525042
Date:		Duration:	60 minutes

Code of Conduct

I 192525042 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Mohammad Sohan

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25

Assignment Title: Agile Sprint Planning & User Story Workshop

Case Study 35 – SDG 12: Sustainable Food Waste Reduction Platform

Problem Statement

Restaurants, supermarkets, and event organizers frequently dispose of surplus food while nearby charities struggle to meet the needs of vulnerable communities. A nonprofit organization proposes a software platform that connects food donors with registered charities for timely food redistribution.

1. Identify the Project Vision and Stakeholders

Project Vision

To develop an Agile-based Sustainable Food Waste Reduction Platform that efficiently connects food donors with registered charities, reducing food waste and ensuring timely food distribution.

Objectives

- Reduce food waste through digital food donation.
- Connect donors with nearby charities.
- Ensure quick and safe food redistribution.
- Provide real-time donation management.
- Support SDG 12 – Responsible Consumption and Production.

Key Stakeholders

- Restaurants
- Supermarkets
- Event Organizers
- Registered Charities / NGOs
- Platform Administrator
- Delivery Volunteers
- Beneficiaries (People in Need)

2. Create Epics and User Stories

Epic 1: User Registration & Login

User Story 1

As a food donor, I want to register an account so that I can donate surplus food.

User Story 2

As a charity, I want to create an account so that I can receive food donations

Epic 2: Food Donation Management

User Story 3

As a donor, I want to add food donation details so that charities can view available food.

User Story 4

As a donor, I want to update or remove food details so that only accurate information is displayed.

Epic 3: Food Request Management

User Story 5

As a charity, I want to request available food so that I can distribute it to needy people.

User Story 6

As a charity, I want to track my request so that I know the delivery status.

Epic 4: Administration

User Story 7

As an administrator, I want to verify donors and charities so that only authorized users use the platform.

3. Define Acceptance Criteria

User Story 1 – Donor Registration

Given

The donor is on the registration page.

When

The donor enters valid details and clicks Register.

Then

The account should be created successfully.

User Story 3 – Add Food Donation

Given

The donor is logged into the platform.

When

The donor enters food details and submits them.

Then

The donation should appear in the available food list.

User Story 5 – Food Request

Given

Food is available for donation.

When

The charity submits a food request.

Then

The donor receives the request and the request status becomes Pending.

4. Prioritize the Product Backlog

Product Backlog Item	Priority	Business Value
User Registration	High	High
User Login	High	High
Add Food Donation	High	High
View Available Food	High	High
Request Food	High	High
Track Donation Status	Medium	Medium
Notifications	Medium	Medium
Reports & Analytics	Low	Low

5. Plan the Sprint Backlog

Sprint Duration: 2 Weeks

User Story	Development Tasks
Register User	Create registration page, Database, Validation
Login	Authentication, Password Encryption
Add Food Donation	Create donation form, Save food details
View Food	Display available food list
Request Food	Request page, Update request status

6. Estimate Effort Using Story Points

Estimation Technique: Fibonacci Sequence (1, 2, 3, 5, 8, 13)

User Story	Story Points	Justification
Register User	3	Simple registration form
Login	3	Basic authentication
Add Food Donation	5	Form validation and database
View Available Food	5	Display and search functionality
Request Food	8	Request processing and status updates
Track Request	5	Request tracking
Notifications	3	Email/SMS notifications

7. Present the Sprint Goal and Expected Deliverables

Sprint Goal

Develop the minimum working product (MVP) that allows users to register, log in, donate food, browse available donations, and request food using Agile Sprint Planning.

Expected Deliverables

- User Registration Module
- User Login Module
- Food Donation Module
- Food Listing Module
- Food Request Module
- Database Connectivity
- Sprint-1 Working Prototype

Agile Sprint Planning

Sprint 1 (Week 1–2)

- User Registration
- User Login
- Add Food Donation
- View Food Donations
- Request Food

Sprint 2 (Week 3–4)

- Donation Tracking
- Notifications
- Charity Dashboard
- Donor Dashboard

Sprint 3 (Week 5–6)

- Reports & Analytics
- Admin Verification
- Performance Improvements
- Final Testing & Deployment

Conclusion

- Agile Sprint Planning divides the project into short development cycles.
- User stories clearly describe user requirements.
- Product backlog helps prioritize features based on business value.
- Story points estimate effort using the Fibonacci method.
- The platform reduces food waste by efficiently connecting donors with charities, supporting SDG 12: Responsible Consumption and Production.